01 – What is AI?

In one sentence: An AI chatbot (technically: an *LLM*, a “Large Language Model”) is a program that produces text by predicting, word by word, what most likely comes next.

> **Remember:** Sounding convincing does not mean being right.

Key points

- **Keep the terms apart:** *AI* is the umbrella term for programs that solve tasks that would otherwise require human thinking. *Machine learning* means: the program learns from many examples instead of being given fixed rules. An *LLM* is one form of that, specialized in language.
- **Text prediction instead of knowledge:** Think of the word suggestions on your phone, only far more powerful. The LLM works with probabilities. It does not check whether something is true.
- **Training vs. use:** During *training*, the model reads huge amounts of text once. After that, it has a knowledge cutoff. Your chat is the *use*: you ask questions, the model answers. It does not automatically know about new events.
- **Strengths:** drafting, summarizing, rephrasing, collecting ideas, writing code.
- **Weaknesses:** facts, current events, exact math, source citations.
- **Chatbot ≠ search engine ≠ human:** A search engine finds existing pages. A chatbot produces a new text. A human has experience and carries responsibility. Some chatbots have web search as an extra feature.

Example: planning a day trip

Risky:

```
Until what time is the outdoor pool in town X open today?
```

Without web search, the AI may give a plausible but made-up time.

Better:

```
Help me plan a day trip to the outdoor pool in town X. You probably
↪ don't know the opening hours for sure. Tell me what you don't
↪ know, and list what I need to verify myself with the outdoor pool.
```

Do / Don’t

Do	Don’t
Use it for texts, ideas and drafts	Blindly rely on it for current facts
Check important details yourself	Take numbers, quotes and sources at face value
Let the AI say when it doesn’t know something	Confuse a confident tone with correctness

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Safety note

Don’t enter sensitive data (passwords, ID or bank details, confidential information). More on this in cheatsheet 04.

Mini checklist

- ☐ I can tell AI, machine learning and LLM apart.
- ☐ I know: an LLM predicts text, it doesn’t “know” anything for sure.
- ☐ I can name two strengths and two weaknesses each.
- ☐ I check facts before passing them on.

Exercise (5 min.)

Ask the same question about a current event twice: once with web search, once without.

```
What happened this week in [your city or your country]? Name three
↪ events.
```

- Does your service have a switch for web search? Test both variants.
- No switch? Pay attention to whether sources or links appear and whether the AI mentions its knowledge cutoff.
- Note down: What was different? How much did you trust the made-up answers?

Next

- Extra exercises: `02_extra_exercises/01_extra_exercises`
- Test: Test A (`03_tests/test_A_foundation`)
- Related: cheatsheet 02 (Tokens & context window), 08 (Hallucinations & fact-checking)

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